# PK-PD Practice 1

Module 7b - PK-PD Relationship · Daily practice sheet · No date

Name _____ Date _____

A certain antibiotic that follows a one-compartment open model has an apparent volume of distribution of approximately 15 L and a first-order elimination half-life of approximately 5 hours and is characterized by a minimum effective concentration of 10 mcg/mL.

**a** What is the teff for a single 500-mg IV bolus dose of this antibiotic?

**b** What is the new teff or for this drug if the dose were increased by a factor of four to 2000 mg?

# Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

## a

Her key

**Answer:** 8.69 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$V_D$** | approximately 15 L | divides the dose: $C_0$ = 33.33 mg/L |
| **t½** | approximately 5 hours | k = 0.1386 $hr^{-1}$ |
| **Ceff** | 10 mcg/mL | equals 10 mg/L, the MEC |
| **$D_0$** | 500-mg | the numerator of $C_0$ |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1};\ Ceff = 10\ mcg/mL = 10\ mg/L$$

Her key uses the k form of the duration line, so the 5-hour half-life is turned into k first. 1 mcg/mL equals 1 mg/L, so the MEC is 10 mg/L.

ALGEBRA

$$C_0 = \frac{500\ mg}{15\ L} = 33.33\ mg/L;\ teff = \frac{\ln(33.33 \div 10)}{0.1386\ hr^{-1}} = \frac{\ln 3.333}{0.1386} = \frac{1.204}{0.1386\ hr^{-1}}$$

$C_0$ is the dose spread through $V_D$. The level only has to fall from 33.33 to 10 mg/L, a 3.33-fold fall, before the effect ends.

ROUND

$$teff = 8.69\ hr$$

1.204 ÷ 0.1386 $hr^{-1}$ = 8.69 hr, her boxed value: between one and two half-lives of 5 hr, because 3.33 lies between 2 and 4.

> **DOES THE NUMBER MAKE SENSE?**
> $C_0$ ÷ Ceff = 3.333 is between 2 and 4, so teff is between one and two half-lives of 5 hr: 8.69 hr.

## b

Her key

**Answer:** 18.69 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 2000 mg | four times 500 mg, so $C_0$ is four times larger |
| **$V_D$** | approximately 15 L | unchanged by the dose |
| **t½** | approximately 5 hours | k = 0.1386 $hr^{-1}$ |
| **Ceff** | 10 mcg/mL | equals 10 mg/L |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{2000\ \text{mg}}{15\ \text{L}} = 133.3\ \text{mg/L};\ k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1};\ C_{eff} = 10\ \text{mg/L}$$

Four times the dose gives four times $C_0$, because $V_D$ does not change with the dose; k comes from the 5-hour half-life and 10 mcg/mL equals 10 mg/L.

ALGEBRA

$$t_{eff} = \frac{\ln(133.3 \div 10)}{0.1386\ \text{hr}^{-1}} = \frac{\ln 13.33}{0.1386} = \frac{2.590}{0.1386\ \text{hr}^{-1}} = 18.69\ \text{hr}$$

The same duration line with the new $C_0$. Four times the ratio adds ln 4 = 1.386 to the logarithm, which is two half-lives once divided by k.

ROUND

$t_{eff}$ = 18.69 hr; change = 18.69 − 8.69 = 10 hr = 2 t½

Her key: the change is 10 hr, two half-lives of 5 hr. A factor of four is two doublings of the dose, and each doubling adds one half-life.

**DOES THE NUMBER MAKE SENSE?**

Four times the dose adds two half-lives, 10 hr, to the 8.69 hr of part a: 18.69 hr, not four times 8.69.